

EXPERIMENT 5

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EXPERIMENT: Advanced SQL Queries

AIM:

To understand and apply advanced SQL techniques, including conditional statements, aggregate functions, different types of joins, self-joins, and date-based operations, for efficiently solving practical database problems.

Question 1:

(i) (A) A website maintains records of users who have registered on the platform and the status of confirmation requests sent to them. A confirmation request can either be successfully completed or may expire without completion.

For each registered user, calculate the confirmation rate, which is defined as:

$$\text{Confirmation Rate} = \frac{\text{Number of Successful Confirmations}}{\text{Total Confirmation Requests}}$$

If a user has never received any confirmation request, their confirmation rate should be considered 0.

Display the user ID along with the confirmation rate rounded to 2 decimal places. The result can be returned in any order.

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Solution:

```
select s.user_id,
round(
avg(
case
when action = 'confirmed' then 1
else 0
end
),2
) as rate
from signups as s left join confirmations as c
on s.user_id = c.user_id
group by s.user_id
order by s.user_id asc;
```

43 Signups AS S LEFT JOIN CONFIRMATIONS

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1

	user_id [PK] integer	confirmation_rate numeric
1	2	0.50
2	6	0.00
3	3	0.00
4	7	1.00

Question 2:

A ride-booking company maintains information about trip requests and the users involved in those trips. Some users may be restricted from using the platform, while others are active participants. Each trip can either be completed successfully or cancelled by one of the participants. For each day between 2013-10-01 and 2013-10-03, determine the proportion of canceled trips among all valid trips. A trip should be considered valid only when both the customer and the driver associated with that trip are active users. The final result should display the date and the calculated cancellation rate rounded to two decimal places.

We need to calculate:

Cancellation Rate =

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Cancelled Trips

Total Valid Trips

But only consider trips where: Client is NOT banned

AND

Driver is NOT banned

And Date between

2013-10-01

And

2013-10-03

Solution:

```
SELECT
  T.REQUEST_AT AS DAY,
  ROUND(
    AVG(
      CASE
        WHEN T.STATUS IN (
          'cancelled_by_driver',
          'cancelled_by_client'
        )
        THEN 1
        ELSE 0
      END
    ), 2
  ) AS "Cancellation Rate"
FROM TRIPS AS T
JOIN USERS AS C
  ON C.USERS_ID = T.CLIENT_ID
JOIN USERS AS D
  ON D.USERS_ID = T.DRIVER_ID
WHERE
  C.BANNED = 'No'
  AND D.BANNED = 'No'
  AND T.REQUEST_AT BETWEEN '2013-10-01' AND '2013-10-03'
GROUP BY T.REQUEST_AT;
```

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Data Output Messages Notifications		
Showing rows: 1 to 3		
	day date	Cancellation Rate numeric
1	2013-10-01	0.33
2	2013-10-02	0.00
3	2013-10-03	0.50

LEARNING OUTCOMES:

1. Learned how to determine confirmation rates by using aggregate functions such as AVG() along with CASE conditions.
2. Understood how LEFT JOIN can be used to display every registered user, including users who do not have any confirmation requests.
3. Learned how self-joins can be applied to the Users table to identify clients and drivers involved in trips separately.
4. Gained experience in finding cancellation rates by combining conditional statements with AVG() and ROUND() functions to get precise results.
5. Improved my ability to combine different SQL concepts such as JOIN, WHERE, date filtering, GROUP BY, ORDER BY, CASE, and aggregate functions to handle practical database-related problems.